SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING

CSA47 – DEEP LEARNING

ASSIGNMENT – 1 : SOLUTIONS

CO1 – Mathematical Foundations and Machine Learning Principles

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Scenario: Healthcare risk prediction. A hospital wants to predict disease risk from age, blood pressure (BP), glucose, cholesterol and family history, using both labelled and unlabelled patient records. All answers below use small worked examples so that every number can be checked by hand.

Task 1 – Patient Data Representation and Normalization

Data used for the example

Three patients are taken, each with four features. Together they form a 3 × 4 feature matrix X, where each row is a patient and each column is a feature (Age in years, BP in mm Hg, Glucose in mg/dL, Cholesterol in mg/dL).

Patient	Age	BP	Glucose	Cholesterol
P1	45	130	110	200
P2	60	150	140	240
P3	30	120	90	180

X = [45 130 110 200 ; 60 150 140 240 ; 30 120 90 180] (3 × 4)

a) Effect of using raw features without centering or normalization

The four columns live on very different scales. Age runs from about 30 to 60, while cholesterol runs from 180 to 240 and the glucose spread is 50. If these are fed in as they are, the feature with the biggest numbers starts to dominate the calculation, simply because of its units and not because it matters more medically.

- **Distance-based methods** (k-NN, K-means) get distorted. The raw distance between P1 and P2 is $\sqrt{(15^2 + 20^2 + 30^2 + 40^2)} = \sqrt{3125} \approx 55.9$. Cholesterol ($40^2$=1600) and glucose ($30^2$=900) contribute 80% of the squared distance, while age adds only 7%.
- **Gradient-based methods** (logistic regression, neural networks) see an uneven loss surface. A weight attached to a large-valued feature changes the output a lot for a tiny update, while a weight on a small-valued feature barely moves. The contours become long thin ellipses, so gradient descent zig-zags and needs many more iterations, or diverges at a learning rate that would have been fine for the other features.
- **No centering** means features are not around zero. Raw values are all large and positive, so the gradients for the weights share the same sign and the bias and weights get entangled. Optimisation becomes slower and numerically less stable.
- **Interpretability** also suffers because weights cannot be compared directly when features are in different units.

b) Feature matrix and normalization procedure

Min–max normalization is used here because it keeps every feature in [0, 1] and is easy to verify by hand. Each column j is rescaled as:

$$x' = (x - min_j) / (max_j - min_j)$$

Feature	Min	Max	Range
Age	30	60	30
BP	120	150	30
Glucose	90	140	50
Cholesterol	180	240	60

Worked calculations for P1: Age = (45−30)/30 = 0.500; BP = (130−120)/30 = 0.333; Glucose = (110−90)/50 = 0.400; Cholesterol = (200−180)/60 = 0.333. P2 holds the maximum of every feature so it becomes 1 everywhere, and P3 holds the minimum of every feature so it becomes 0 everywhere.

Patient	Age′	BP′	Glucose′	Chol′
P1	0.500	0.333	0.400	0.333
P2	1.000	1.000	1.000	1.000
P3	0.000	0.000	0.000	0.000

Numerical verification: every column minimum is 0 and every column maximum is 1, so all features now share the same range. Redoing the P1–P2 distance, each feature difference is now at most 0.667, so no single feature swamps the others.

Alternative (z-score standardization). With $z = (x - \mu)/\sigma$ (sample standard deviation), the means are $\mu = (45, 133.33, 113.33, 206.67)$ and $\sigma = (15, 15.28, 25.17, 30.55)$. This gives:

Patient	Age z	BP z	Glucose z	Chol z
P1	0.000	−0.218	−0.132	−0.218
P2	1.000	1.091	1.060	1.091
P3	−1.000	−0.873	−0.927	−0.873

Verification: each column sums to ≈ 0 (e.g. glucose: $-0.132 + 1.060 - 0.927 \approx 0$) and has standard deviation 1. Z-scores are preferred when outliers exist, since min–max is sensitive to a single extreme value.

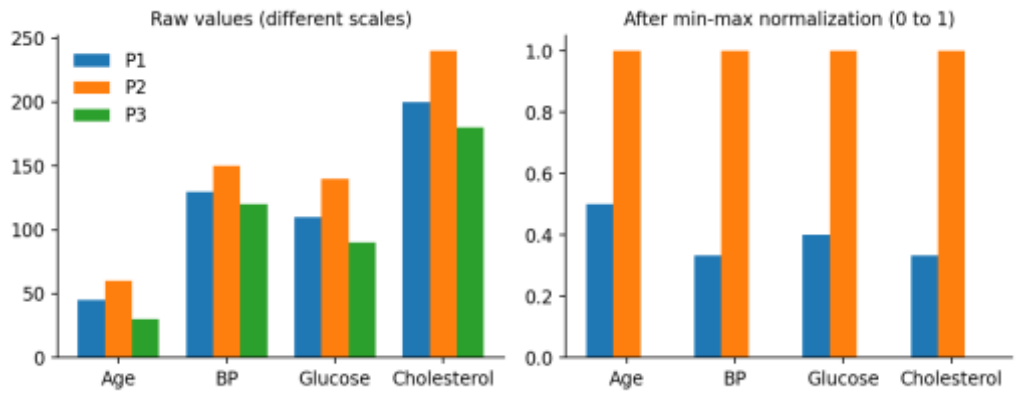


Figure 1: The three patients before and after min–max normalization

c) Why normalization matters for distance-based and gradient-based methods

- **Distance-based:** Euclidean distance adds squared differences across features. Unless the features are on a comparable scale, the one with the largest numeric range decides who counts as a "nearest neighbour" or which cluster a patient falls in. Normalization makes each feature contribute fairly.
- **Gradient-based:** for a loss L(w), the gradient with respect to weight w_j is proportional to the feature x_j. If x_j is large, its gradient is large. Normalized inputs make the loss surface closer to circular, so one learning rate suits all weights and convergence is faster and steadier.
- It also prevents numerical overflow in exponentials (sigmoid, softmax) and keeps activations in a sensible range for deep networks.

Interpretation: after normalization P1 is a mid-range patient (about 0.33 to 0.5 on each scale), P2 is at the high end of everything and P3 at the low end. These values are now comparable across features.

Task 2 – Vector Representation and Matrix Operations

a) Vector representation and weighted score

Let the feature vector of one patient be $x = [x_1, x_2, x_3, x_4]^T$ for (age, glucose, BP, cholesterol), and the importance weights be $w = [w_1, w_2, w_3, w_4]^T$. The preliminary risk score is the dot product:

$$s = w^T x = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

For all patients at once, with X as the n × 4 matrix, the score vector is s = Xw, an n × 1 vector with one score per patient.

b) Numerical example

Take the normalized values of P1 from Task 1: $x = [0.50, 0.40, 0.333, 0.333]^T$. The hospital gives these weights: age 0.30, glucose 0.40, BP 0.20, cholesterol 0.10 (they add up to 1, so the score stays between 0 and 1).

$$s = (0.30)(0.50) + (0.40)(0.40) + (0.20)(0.333) + (0.10)(0.333)$$

$$s = 0.150 + 0.160 + 0.067 + 0.033 = 0.41$$

Using the matrix form on all three normalized patients, $Xw = [0.41, 1.00, 0.00]^T$ for P1, P2 and P3.

Interpretation: P1 has a moderate preliminary risk of 0.41. P2 scores 1.0 because every feature is at its highest, and P3 scores 0. Glucose has the biggest weight, so a change in glucose moves the score more than an equal change in cholesterol. This is only a screening score; the weights would normally be learned from data.

Task 3 – Probability and Conditional Probability

Event definitions

- D = patient has the disease; D^c = patient does not have the disease
- E = patient has elevated glucose

	Elevated glucose (E)	Normal glucose	Total
Disease (D)	160	40	200
No disease (D^c)	120	680	800

Total	280	720	1000

a) P(E)

$$P(E) = (160 + 120) / 1000 = 280/1000 = 0.28$$

Equivalently by total probability: $P(E) = P(E|D)P(D) + P(E|D^c)P(D^c) = (0.8)(0.2) + (0.15)(0.8) = 0.16 + 0.12 = 0.28$.

b) P(D | E)

$$P(D|E) = P(D \cap E) / P(E) = (160/1000) / (280/1000) = 160/280 \approx 0.571$$

c) P(E | D)

$$P(E|D) = P(D \cap E) / P(D) = 160/200 = 0.80$$

Check with Bayes: $P(D|E) = P(E|D)P(D)/P(E) = (0.8)(0.2)/0.28 = 0.571$. ✓

d) Use in prioritizing patients for further testing

$P(E|D) = 0.80$ tells us the sign is common in sick patients, but $P(D|E) = 0.571$ is the number that matters in the clinic, because the doctor sees the glucose result and wants to know the chance of disease. The base rate for the whole sample is only 20%, so an elevated glucose reading raises the probability almost three times (0.20 → 0.57). Patients with a higher posterior can be placed earlier in the queue for confirmatory tests, and a patient who has several risk indicators can have the conditional probability updated again. Note that $P(E|D)$ and $P(D|E)$ are different quantities, and confusing them is a common mistake in medical reasoning.

Task 4 – Bayesian Statistics for Disease Classification

Given: $P(D) = 0.10$, $P(+|D) = 0.90$ and $P(+|D^c) = 0.05$. So $P(D^c) = 0.90$.

a) Posterior using Bayes' theorem

$$P(D|+) = P(+|D)P(D) / [\, P(+|D)P(D) + P(+|D^c)P(D^c) \,]$$

Step 1 (true positives): $0.90 \times 0.10 = 0.09$

Step 2 (false positives): $0.05 \times 0.90 = 0.045$

Step 3 (evidence P(+)): $0.09 + 0.045 = 0.135$

$$P(D|+) = 0.09 / 0.135 = 0.667 \text{ (about 66.7\%)}$$

Sanity check with 1,000 people: 100 have the disease and 90 test positive; of the 900 healthy people, 45 test positive. So 90 out of 135 positives are truly diseased = 66.7%. ✓

b) Roles of prior and likelihood

- **Prior P(D)** is what we believe before seeing the test: the disease prevalence of 10%. A low prior means that even a good test produces many false positives, because there are so many more healthy people.
- **Likelihood P(+|D)** (together with $P(+|D^c)$) says how strongly the observed result favours the disease over no disease. Posterior $\propto$ likelihood × prior.
- The posterior balances both. Here the test is quite accurate (90% sensitivity), yet the posterior is only 67% since the prior is small. If the prior were 50%, the same test would give 0.45/(0.45+0.025) ≈ 94.7%.

c) Why Bayesian reasoning helps when medical information is uncertain

- It combines prior knowledge (prevalence, earlier results, expert opinion) with new evidence in a consistent way.
- It outputs a probability, not a hard yes/no, so doctors can see how confident the system is.
- It can be updated sequentially: today's posterior becomes tomorrow's prior when a second test result arrives.
- It works with small or noisy data and handles missing information by averaging over possibilities.

Task 5 – Entropy and Information Gain

a) Entropy of the risk classification

Of 100 records, p(high) = 0.4 and p(low) = 0.6. Entropy is $H = -\Sigma\, p_i \log_2 p_i$.

$$H = -0.4\ log_2(0.4) - 0.6\ log_2(0.6)$$

$$H = 0.4(1.3219) + 0.6(0.7370) = 0.5288 + 0.4422 = 0.971\ bits$$

b) What the entropy says about uncertainty

The maximum entropy for two classes is 1 bit (a 50/50 split) and the minimum is 0 (all records in one class). A value of 0.971 is very close to the maximum, so the dataset is highly mixed and a randomly chosen record is hard to predict without more information. In other words, about 0.97 bits of uncertainty remain to be removed by features.

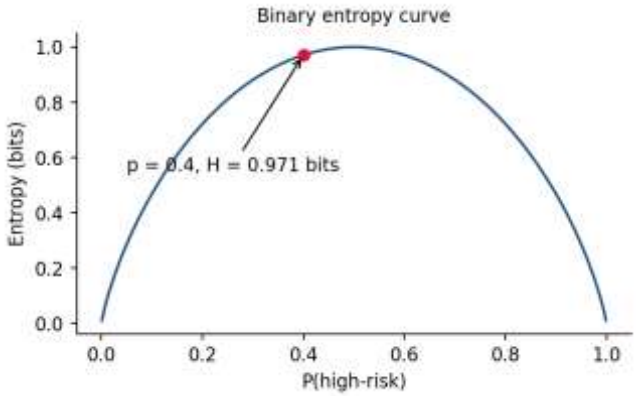


Figure 2: Binary entropy; the current dataset sits near the peak

c) Information gain and feature selection

$$IG(S, A) = H(S) - \Sigma\ (\ |S_v| / |S|\ )\ H(S_v)$$

Information gain is the drop in entropy after splitting the data on a feature. A feature with large gain separates high-risk and low-risk patients well, so it is a good one to pick (this is how decision trees choose splits).

Illustration with glucose (assumed split for demonstration): suppose 50 patients have high glucose, of whom 35 are high-risk, and 50 have normal glucose, of whom 5 are high-risk. (Totals: 40 high-risk, 60 low-risk, matching the dataset.)

- H(high glucose) = $-0.7 \log_2 0.7 - 0.3 \log_2 0.3 = 0.881$
- H(normal glucose) = $-0.1 \log_2 0.1 - 0.9 \log_2 0.9 = 0.469$
- Weighted entropy = 0.5(0.881) + 0.5(0.469) = 0.675
- **IG(glucose) = 0.971 − 0.675 = 0.296 bits**

The gain of about 0.30 bits means that knowing the glucose group cuts roughly 30% of the uncertainty. If another feature gave a lower gain, glucose would be chosen first.

Task 6 – Numerical Computation and Distance Measurement

a) Euclidean distance

Patient A = (0.70, 0.80, 0.60) and Patient B = (0.50, 0.75, 0.90) for (glucose, BP, cholesterol).

$$d(A,B) = \sqrt{[\ (a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2\ ]}$$

Feature	A	B	Difference	Squared
Glucose	0.70	0.50	0.20	0.0400
BP	0.80	0.75	0.05	0.0025

Cholesterol	0.60	0.90	−0.30	0.0900
Sum				0.1325

$$d(A,B) = \sqrt{0.1325} \approx 0.364$$

b) Distance as similarity

A small distance means two patients have similar feature profiles, and a large distance means they are different. Since the largest possible distance in this 3-feature unit cube is $\sqrt{3} \approx 1.73$, a distance of 0.364 shows that A and B are fairly close, though not identical. Most of the gap comes from cholesterol (0.09 of 0.1325). This is the basis for k-NN (look at the nearest patients) and for clustering.

c) Why normalization is necessary

If glucose was in mg/dL (about 100), BP in mm Hg (about 130) and cholesterol in mg/dL (about 200), the differences would be dozens of units and be dominated by whichever feature has the widest numeric range. Changing the unit, say cholesterol from mg/dL to mmol/L, would change the distance although the patients are unchanged. Normalization removes this unit dependence so that every feature has comparable influence, and the distance then reflects clinical similarity.

Task 7 – Bias–Variance Trade-off

a) Identification

Model	Training error	Validation error	Diagnosis
A	High	High	High bias (underfitting)
B	Very low	High	High variance (overfitting)
C	Low	Low, close to training	Good balance

b) Effect of model complexity

- **Low complexity:** the model is too rigid (e.g. a straight line for a curved relationship). It makes systematic errors, so bias is high and variance is low.
- **High complexity:** the model is flexible enough to fit even the noise in the training sample. Bias falls but variance rises, because a different training sample would produce a very different model.

Expected error = Bias² + Variance + Irreducible noise. As complexity grows, bias² decreases and variance increases, so total error is U-shaped and the best model sits near the bottom of the U (Model C).

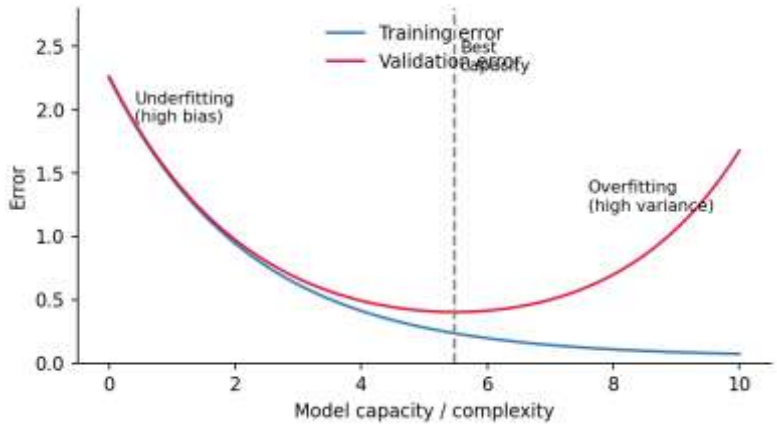


Figure 3: Training and validation error against model complexity

c) Improvements

- **Model A (high bias):** increase capacity (more features, polynomial or interaction terms, more layers/units), reduce regularization, train longer, or add clinically informative features such as family history.
- **Model B (high variance):** collect more data, add L1/L2 regularization or dropout, simplify the model, use early stopping, cross-validation, feature selection or ensembling (bagging).

Task 8 – Overfitting, Underfitting and Model Capacity

a) Identification

- Simple model that fails to capture patterns → **underfitting**.
- Moderately complex model that performs consistently → **appropriate generalization**.
- Highly complex model, near-perfect on training but poor on unseen patients → **overfitting**.

b) Capacity and learning complex relationships

Capacity is the range of functions a model can represent (for instance, the degree of a polynomial, or the number of layers and neurons). A low-capacity model, such as a linear classifier, can only draw a simple boundary, so it cannot represent interactions like “high glucose AND family history is far riskier than either alone”. More capacity allows curved and interacting decision boundaries and hence more complex disease patterns.

c) Why more capacity first helps and later hurts

At the start, the model is too limited, so extra capacity lets it capture real structure in the data and both training and validation errors fall. Once the real pattern has been captured, the remaining capacity is spent fitting the random noise and quirks of the particular training patients. Training error keeps going down but validation error turns upward, and the gap between them (the generalization gap) widens. This is why the best capacity is picked using validation data (Figure 3), and why the amount of data matters: more records support a higher-capacity model before overfitting begins.

Task 9 – Supervised Learning and Model Evaluation

a) Why it is supervised

Each historical record has input features (age, BP, glucose, cholesterol, …) and a known label (diseased or disease-free). The model learns a mapping f: X → y from labelled examples and then predicts the label of new patients. Because the target is a discrete category, it is a supervised classification problem.

b) Roles of training and testing sets

- **Training set:** used to fit the model parameters (and a validation part is used to tune hyperparameters).
- **Testing set:** kept aside and used only once at the end to estimate how the model performs on unseen patients. This gives an honest measure of generalization, since the model could simply memorise the training data.

c) Accuracy

$$Accuracy = correct\ predictions / total\ predictions = 92 / 100 = 0.92 = 92\%$$

The model classifies 92 of every 100 new patients correctly and makes 8 errors. This looks good, but accuracy alone can mislead in healthcare. If only 8% of patients had the disease, a model that always says “disease-free” would also score 92%. So precision, recall (sensitivity) and the confusion matrix should be examined too, because missing a sick patient (false negative) is usually more costly than an extra test.

Task 10 – Unsupervised Learning and Patient Grouping

a) Why unsupervised learning

There are no risk labels, so there is no target to predict. The goal becomes finding structure in the data itself, which is what unsupervised learning does. Clustering can reveal natural patient groups, such as a high-glucose/high-cholesterol group or a young low-risk group, without anyone labelling thousands of records by hand.

b) K-means procedure

- Normalize all features and choose K (using the elbow method or silhouette score).
- Pick K initial centroids (e.g. K-means++).
- **Assignment step:** assign each patient to the nearest centroid.
- **Update step:** recompute each centroid as the mean of its assigned patients.
- Repeat until assignments stop changing; the objective $\Sigma \|x_i - \mu_c(i)\|^2$ keeps decreasing.

Small example (normalized glucose, BP): P1 (0.1, 0.2), P2 (0.2, 0.1), P3 (0.8, 0.9), P4 (0.9, 0.8), K = 2 with initial centroids P1 and P3. P2 is 0.141 from centroid 1 but about 1.1 from centroid 2, so it joins cluster 1; P4 joins cluster 2 in the same way. New centroids: (0.15, 0.15) and (0.85, 0.85). A second pass changes no assignments, so the algorithm has converged.

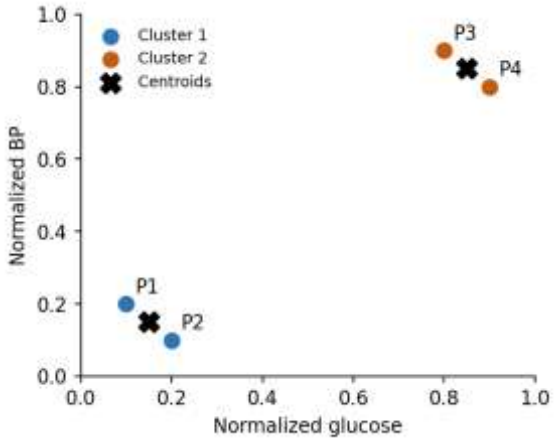


Figure 4: K-means result on the four-patient example

c) Distance as similarity

K-means uses the Euclidean distance $d(x, \mu) = \sqrt{\Sigma (x_j - \mu_j)^2}$. Patients close to one another in feature space are alike and fall into the same cluster; distant ones are put in different clusters. This is why normalization has to be done before clustering.

d) Clinical use without replacing diagnosis

The clusters are only hypotheses. Doctors can study each group's average profile, notice a group with high glucose and cholesterol and invite it for screening, or look at atypical patients far from every centroid. Cluster membership can also help plan resources and design follow-up programmes. However, clusters carry no official disease labels and K-means can group patients for reasons unrelated to health, so every conclusion must be confirmed by clinical tests and a qualified doctor, who makes the diagnosis.

Task 11 – Comparison of Supervised and Unsupervised Learning

a) Suitable paradigm for each dataset

- **Dataset A (features + disease labels):** supervised learning, as we have the correct answers to learn from (classification).
- **Dataset B (features only):** unsupervised learning (clustering, dimensionality reduction), since there is no target variable.

b) Comparison

Aspect	Supervised (Dataset A)	Unsupervised (Dataset B)
Data	Labelled	Unlabelled
Objective	Learn mapping from features to label; minimize prediction error	Discover hidden structure or groups
Output	Predicted class or probability of disease	Cluster assignments, centroids, low-dimensional representation
Evaluation	Accuracy, precision, recall, F1, AUC on a test set	Silhouette, within-cluster variance, clinical review
Examples	Logistic regression, decision tree, neural network	K-means, hierarchical clustering, PCA

c) Recommendation for predicting disease status of a new patient

Use **supervised classification trained on Dataset A**, because the task is a direct prediction of a known label and supervised methods can be measured against ground truth. Normalize the features with the training-set statistics and select the capacity through validation. Dataset B can still help: clustering it (and Dataset A) can supply extra features, reveal subgroups and outliers, or provide pseudo-labels in a semi-supervised setting. Unsupervised learning alone should not be used for the final prediction, as clusters do not map reliably to disease status.

Task 12 – Design Challenge: Selecting and Adapting a Learning Strategy

Decision: the original approach (analyse 10,000 records) should be **modified**, not thrown away and not kept unchanged. The core ideas (normalization, probability-based classification, distance-based grouping) still apply, but the pipeline must become scalable, robust to missing data and must use both the labelled and the unlabelled records.

A. Impact Analysis

a) Effect of dataset size (10,000 → 1,000,000)

- **Representation:** the feature matrix grows from 10,000 × d to 1,000,000 × d. With d ≈ 5 and 8 bytes per value this is only about 40 MB, so it still fits in memory, but one-hot or high-dimensional features (e.g. text notes) would not. Sparse storage and float32 reduce memory further.
- **Numerical computation:** costs are linear or worse in n. Computing a full pairwise distance matrix needs $n^2 = 10^{12}$ entries, which is impossible, so we need mini-batch methods, approximate nearest-neighbour search and vectorised matrix operations. K-means costs $O(n \cdot K \cdot d)$ per iteration, which is fine. Repeated additions of many numbers also raise floating-point round-off concerns, so stable computation (e.g. log-sum-exp, float64 accumulators) matters.
- **Learning efficiency:** full-batch gradient descent becomes slow, so mini-batch SGD is used. More data lowers variance and supports a higher-capacity model: the standard error of an accuracy estimate shrinks as $1/\sqrt{n}$ (about 10 times smaller than with 10,000 records). Training and cross-validation time grows, so parallel or distributed processing may be needed.

b) Importance of normalization, missing-value handling and variability

- **Normalization:** features from different hospital systems have different scales; without it, distance and gradient methods behave badly (Tasks 1 and 6). Scaling parameters must be learned from the training data only and then applied to new data, to avoid leakage.
- **Missing values:** with a million records, some blanks are certain (a lab test not done). Dropping every incomplete row would waste data and may bias the sample if the missingness is related to health. Use median/mean imputation, k-NN or model-based imputation, plus a "was missing" indicator.

- **Variability and outliers:** measurement error, different labs and extreme values can inflate the variance and distort means and min–max ranges. Robust scaling (median/IQR), outlier checks and monitoring of distribution shift are required.

B. Evaluation of the Original Approach

a) Is a single supervised approach sufficient? No. A supervised classifier predicts disease risk well when labels exist, but it does not tell us what natural patient groups there are, and the two goals are different: prediction needs a target, group discovery needs structure. A single model would also have to be trained on the labelled part only.

b) Limitations of labelled learning when useful patterns exist in unlabelled records

- Labelling is expensive (needs doctors and follow-up), so most of the 1,000,000 records are probably unlabelled and would be ignored.
- A small labelled sample may not represent all patient types, which leads to bias and poor generalization on rare subgroups.
- Labels can be noisy or delayed, and a supervised model can only learn the categories it was given; it cannot spot a new, unexpected patient group.
- The structure in the unlabelled data (density, clusters) could improve the decision boundary but is wasted.

C. Adaptation or Redesign

a) Proposed strategy: a combined, semi-supervised pipeline

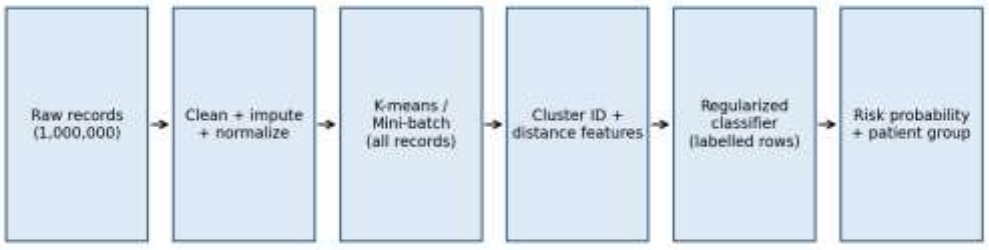


Figure 5: Proposed combined unsupervised + supervised pipeline

- **Step 1 – Data preparation:** clean, impute missing values, add missing indicators, robust-scale numeric features using training statistics.
- **Step 2 – Unsupervised stage:** run (mini-batch) K-means on all 1,000,000 records to discover patient groups; choose K by silhouette/elbow and clinical review.

- **Step 3 – Feature enrichment:** add cluster ID and distance-to-centroid as features for each patient.
- **Step 4 – Supervised stage:** train a regularized classifier (logistic regression or a small neural network) on the labelled records, with probability outputs for risk.
- **Step 5 – Optional semi-supervised step:** assign confident pseudo-labels to unlabelled records, retrain and compare.
- **Step 6 – Output:** disease-risk probability plus patient group for each new patient, to be reviewed by clinicians.

b) How the mathematical ideas are incorporated

- **Probability:** the classifier outputs P(disease | x). Bayes' theorem combines the prevalence (prior) with the likelihood, and thresholds can be chosen to prioritise high-posterior patients (Tasks 3 and 4). Missing values can be handled with probabilistic imputation.
- **Distance measures:** Euclidean distance on normalized features drives K-means and the distance-to-centroid features; it can also detect outliers.
- **Normalization:** applied before every distance or gradient calculation so no feature dominates; also speeds up SGD.
- **Model capacity:** with 100 times more data, capacity can be increased (more features, interactions, a deeper network) but controlled by L2 regularization, dropout, early stopping and validation curves, so the model stays at the sweet spot between underfitting and overfitting (Tasks 7 and 8).

D. Evidence-Based Defence

a) Experiments and evaluation measures

Experiment	What is compared	Measures
E1 – Baseline	Original supervised-only model on labelled data	Accuracy, precision, recall, F1, ROC-AUC
E2 – Redesigned	Clustering features + supervised classifier	Same measures on the same hold-out set
E3 – Semi-supervised	Add pseudo-labelled records	Change in recall/AUC; calibration (Brier score)
E4 – Clustering quality	Different K values, scaled vs unscaled	Silhouette, Davies–Bouldin, within-cluster SS, clinician review

E5 – Scalability	Runtime and memory at 10^4, 10^5, 10^6 records	Training time, memory, cost per prediction
E6 – Robustness	Different imputation methods, added noise	Metric stability across folds

All experiments use stratified k-fold cross-validation with a final untouched test set, identical splits for every method, and a paired test or confidence intervals to see whether improvements are real. Recall is emphasised because a missed patient is costly.

b) Justification of the final design

- **Mathematical reasoning:** Bayes' theorem and calibrated probabilities support risk ranking; distance-based clustering on normalized data is well defined; the bias–variance trade-off justifies adjusting capacity with regularization as data grow.
- **Generalization:** much more data lowers variance, and the unlabelled records add information about the real patient population, so the model is less likely to overfit the small labelled sample. Validation on held-out data confirms this.
- **Computational needs:** K-means is O(nKd) per iteration and mini-batch variants and SGD scale to millions of rows on ordinary hardware; no n^2 computation is required.
- **Data characteristics:** missing values, different scales and mixed labelled/unlabelled data are explicitly handled by imputation, normalization and the two-stage design.

Conclusion: keep the foundations (normalization, probability, distance, capacity control), modify the strategy into a scalable hybrid pipeline, and keep clinicians in the loop for the final decision. The redesigned approach is adopted only if the experiments show it matches or beats the baseline on recall and AUC at an acceptable computational cost.

Summary of Tasks and Marks

Task	Topic	Marks
1	Feature representation and normalization	5
2	Vector representation and matrix operations	2
3	Probability and conditional probability	6
4	Bayes' theorem	4
5	Entropy and information gain	3
6	Distance measurement	3
7	Bias–variance trade-off	4

8	Overfitting, underfitting and capacity	4
9	Supervised learning and evaluation	3
10	Unsupervised learning – clustering	4
11	Comparison of learning approaches	4
12	Design challenge	8
	Total	50